

A healthcare cost proxy from UKHLS alone

What is buildable now, how to validate it, and when linkage becomes worth it

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Companion to `health_measures_note.pdf` §8. That section established that expected cost, not a binary event, is the right outcome for the framework’s convexity question, and that UKHLS carries the physical quantities cost is bought with. This is the plan for turning those quantities into a defensible cost proxy without linkage.

1 What we are building

A per-person-wave estimate of annual NHS secondary-care cost, in a fixed price year, for waves 7–15. It will be wrong in level and, we hope, roughly right in *shape* — which is all the framework needs, since every question we are asking is about how cost varies with health, not about its absolute size. The one-line target:

$$\hat{C}_{it} = \underbrace{c_{\text{spell}}(k) + c_{\text{bed}} \cdot (\text{nights}_{it} - \bar{L})^+}_{\text{in-patient}} + \underbrace{c_{\text{op}} \cdot \hat{n}_{it}^{\text{op}}}_{\text{out-patient}}$$

with k the attributed condition where we have one, and everything indexed to a single price year.

2 Ingredients we already hold

Quantity	Source	Status
Any admission	<code>hosp</code> , waves 7–15	complete for respondents
Nights	<code>hospd</code>	all admitted; median 3, mean 7.5, max 300
Maternity flag	<code>hospch</code>	12.7% of admissions, mean age 31.7
Nights by new condition	<code>hospdci</code> , waves 2–15	21% of admissions
Nights by prior condition	<code>hospdcp</code> , waves 11–15	11% of admissions
either		29% of admissions, 65% of nights
Out-patient band	<code>h12gp</code> , <code>h12hop</code>	banded 0/1–2/3–5/6–10/>10

Unit costs come from the NHS National Cost Collection, which publishes national average unit costs annually and free, at both aggregate and HRG level. Indicative recent figures: non-elective in-patient episode $\approx$ £1,590, elective $\approx$ £3,684, day case $\approx$ £738, excess bed day $\approx$ £313, out-patient attendance $\approx$ £120.

3 Four stages

Stage 1 — a flat-rate baseline (half a day)

Cost each admission at the non-elective episode rate plus excess bed days beyond the average inclusive length of stay, and each out-patient attendance at the flat rate, with band midpoints 0, 1.5, 4, 8, 15. Exclude maternity. This is deliberately crude and exists to give every later stage something to be compared against.

The one real modelling decision here. UKHLS gives total nights but *not the number of spells*, so a person with 20 nights might have had one long admission or four short ones, which differ in cost by a factor of about three. Stage 1 assumes one spell. Stage 4 tests how much that matters; if the answer is “a lot”, it becomes the main argument for linkage, because HES would resolve it exactly.

Stage 2 — condition weighting (one to two days)

For the 65% of nights carrying an attribution, replace the flat rate with a condition-specific one. The mapping is UKHLS condition code → NCC HRG chapter → average cost, at chapter level rather than individual HRG, since the survey identifies a condition and not a procedure. The attributed nights concentrate in stroke, COPD, asthma, osteoarthritis, MI, congestive heart failure and chronic kidney disease, which span a wide cost range, so this is where the proxy should improve most.

Unattributed nights take the case-mix-weighted average of the attributed ones *within the same health decile*, rather than a global average — otherwise the imputation flattens exactly the gradient we are trying to measure.

Stage 3 — validation against published profiles (one day)

This is the stage that decides whether the proxy is usable, and it is the reason to prefer building it here rather than assuming a shape.

1. **The age profile.** The CHE reference tables give average annual NHS spend by age, sex and deprivation, and the IFS has published hospital spending profiles from NHS administrative records. Our proxy should reproduce the shape — flat to about 50, rising steeply after 70 — without having been fitted to it. A proxy that misses the age profile cannot be trusted on the health profile.
2. **The concentration.** Published figures put the top decile of spenders at more than half of expenditure. Ours should land near that. Too little concentration means the intensive margin is being under-weighted; too much means outliers are running the estimate.
3. **The level.** Per-capita secondary care spend implied by the proxy against national totals. We expect to *undershoot*, because UKHLS misses A&E, and self-reported nights are known to be under-recalled; the size of the shortfall is itself informative.

Stage 4 — use it, and test what it rests on (one to two days)

Redo §8’s convexity analysis with $\mathcal{L}$ on the vertical axis, and §11’s outcome prediction with cost as the outcome. Then the sensitivity analysis that matters:

- **Spells.** Re-run assuming one spell, two spells, and nights/ $\bar{L}$ spells. If the convexity conclusion is stable across these, the spell ambiguity is not binding and linkage buys less than it looks.
- **Outlier handling.** The 300-night maximum will dominate any mean. Winsorise at the 99th percentile and re-run.
- **Condition weights.** Flat rate against condition-weighted: does the shape change, or only the level?
- **Maternity.** Already known to move the gradient from $31.9\times$ to $52.7\times$; carry both through.

4 What stages 1–3 actually found

Stages 1–3 are built (`data_cleaning/09_build_cost_proxy.py`, `descriptives/15_cost_validation.py`). The utilisation sample is 301,390 person-waves on 59,750 people, waves 7–15; the validation restricts to ages 20–90, 283,852 person-waves.

A third tier appeared. The plan assumed two tiers of contact. There are three: `h12gp` is “Visited GP in last 12 months” and `h12hop` is “Hosp or clinic out-patient last 12 months”, both banded 0–4 across waves 7–15. An earlier draft of the companion note had these two swapped and asserted that the mainstage carried no GP-visit count; the labels were re-checked directly in waves g, j and o and the note is corrected.

Stage 1. At indicative unit costs the flat index averages £539 per person-year: £243 in-patient, £182 out-patient, £114 GP.

Stage 2. Condition attribution reaches 18.0% of admissions pooled over waves 7–15 — below the 29% measured on wave 15 alone, because `hospcp/hospdcp` only begin at wave 11. Weighting nights by condition moves the mean to £566, about 6%. More useful is what it does *not* do: the imputed condition weight is nearly flat across deciles of physical health, 1.16 to 1.32. Sicker people are not admitted for systematically more expensive conditions — they are admitted more often and for longer. Condition weighting is a level adjustment, not a shape one, which answers one of Stage 4’s sensitivity questions in advance.

Stage 3: two checks pass, one strains

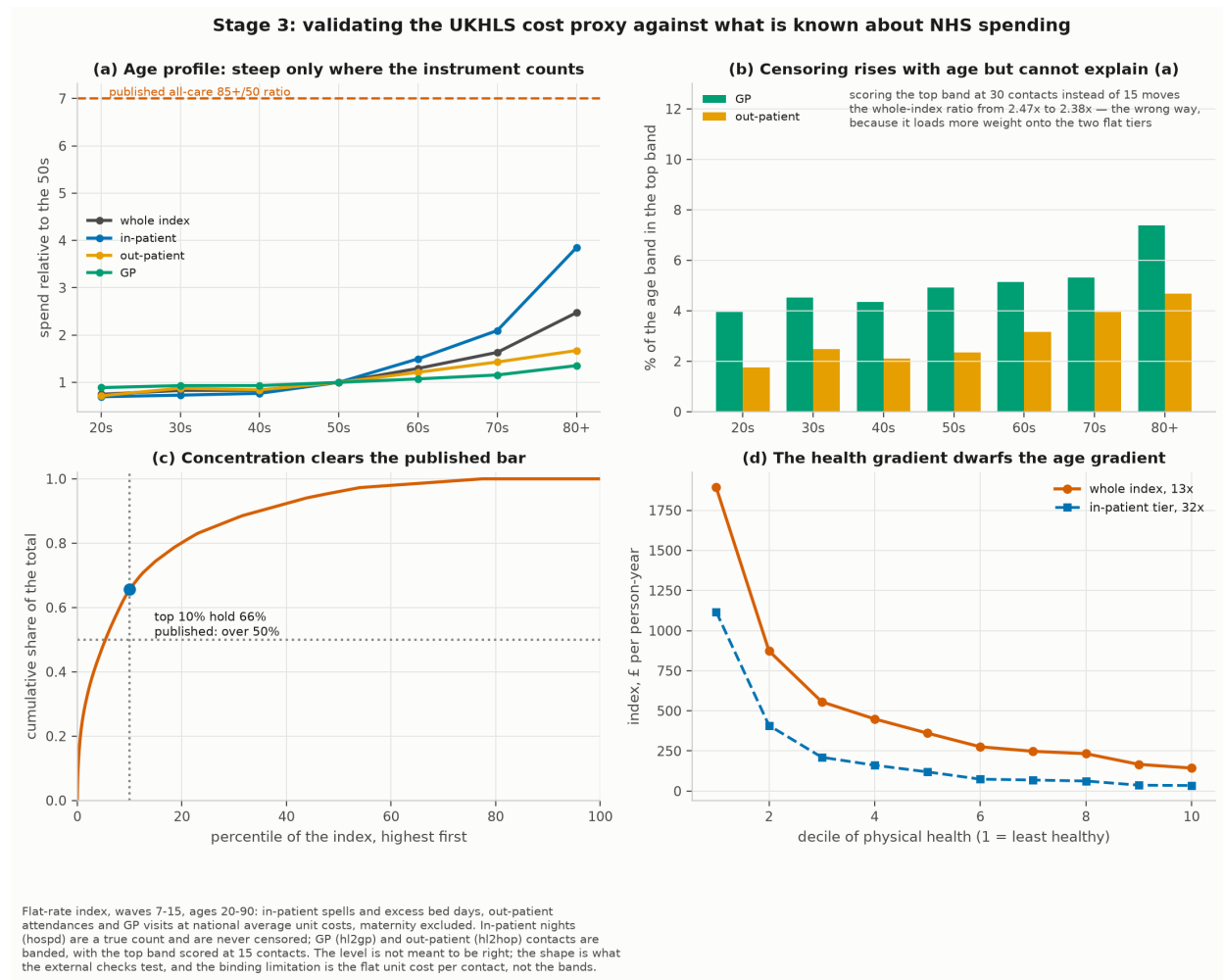


Figure 1: Validation of the flat-rate index against what is independently known about NHS spending. Panel (a) gives the age profile of the whole index and of each tier, relative to the 50s, against the published all-care ratio; (b) the share of each age band top-coded at “more than ten” contacts, with the top-band sensitivity; (c) the concentration curve against the published top-decile share; (d) the gradient in physical health, the quantity the proxy is actually for.

Concentration passes. The top 1% of person-waves hold 25.2% of the total, the top 5% hold 48.5% and the top decile 65.6%, against a published benchmark of more than half. If anything we are over-concentrated, which is expected: 22.7% of person-waves are exactly zero on an index that carries no prescribing, A&E or community contact, and those are the components that would give almost everyone a small positive amount.

Level undershoots, as predicted. £539 per person-year against NHS spend per head of roughly £3,000. This was expected and is not informative — three tiers of contact priced at national averages cannot add to the national total.

The age profile is the one that strains. The whole index puts the 80+ band at $2.5\times$ the 50s, against a published all-care ratio near $7\times$. Decomposed by tier the picture is sharper: in-patient $3.9\times$,

out-patient 1.7 $\times$, GP 1.4 $\times$. The one tier measured as a true count is the one with a credible gradient; the two banded tiers are nearly flat in age.

But the bands are not the reason. Top-coding does rise with age (7.4% of the 80+ band at “more than ten” GP visits against 4.9% of the 50s), so it was the natural suspect. Scoring the top band at 30 contacts rather than 15 moves the whole-index ratio from 2.47 $\times$ to 2.38 $\times$ — the wrong way, because it loads more weight onto the two tiers that are flat in age. The censoring is real and too small to matter.

The binding limitation is the other one: we hold *cost per contact* fixed. An out-patient attendance for an 85-year-old and one for a 30-year-old enter at the same £120, when the real cost differs by specialty and complexity. UKHLS gives us contact counts and no case mix outside the 18% of admissions with an attributed condition. The flat unit cost, not the banded instrument, is what flattens the age profile — and it is exactly what linkage would fix.

Verdict

Usable for what it is for, with one stated caveat. The proxy’s purpose is the gradient in *health*, and there it is strong and robust: 13.2 $\times$ between the least and most healthy decile on the whole index, 32.3 $\times$ on the in-patient tier, against an age gradient of 2.5 $\times$. Age is a poor proxy for the health that drives cost, which is the substantive point the conceptual framework rests on.

The caveat is that the age profile is understated by roughly a factor of three, so the proxy should not be used for anything whose answer depends on the age gradient — lifetime cost projections, discounting, or the timing of prevention returns. For those, either report the in-patient tier alone, or wait for linkage. Results should carry the in-patient tier separately throughout, since it is the only component with both a true count and a credible profile.

5 What stage 4 found

Built as `descriptives/16_cost_convexity.py` and `17_cost_age_interaction.py`; reported in full in `health_measures_note.pdf` §12.

The costing assumptions do not matter; the health scale does. All twelve variants of the sensitivity grid — one spell, two spells, one per five nights; maternity in or out; flat or condition-weighted; top band at 15 or 30 — are convex in levels and in logs, on the physical GRM θ . Winsorising at the 99th percentile halves the magnitude and changes nothing else. The spell ambiguity, which looked like the most damaging gap in the data, is not binding. What does bind is the horizontal scale: with pounds on the vertical axis, θ gives a log-scale quadratic of +.086 and its own TCC gives −.111 on identical people with an identical ranking. Fixing the vertical axis was necessary and is not sufficient.

The health–cost relationship is close to age-invariant. At a fixed θ the cost index is nearly flat in age, shallow-U-shaped with a midlife minimum: the sickest tenth costs £2,138 at 20–34 and £2,112 at 75–90. The sickest-to-healthiest ratio is 10–14 $\times$ at every age. The absolute gradient does rise, £350 to £734 per SD, but on the common support every band populates it runs £438 to £555, and the proportional gradient is flat (0.907 to 0.882 log points). The $\theta \times$ age interaction is insignificant in pounds ($t = 1.2$). Ageing raises cost by moving people down a fixed curve — 2.3% of the 20–34 band sits in the sickest pooled tenth against 24.9% of the 75–90 band — not by changing its shape.

This is the result the whole proxy was built to get at, and it is one the stage 3 weakness does not undermine: the age profile is understated, but the finding is precisely that health rather than age is what carries cost.

6 What this cannot do

Primary care, prescribing, A&E and community services are absent entirely, and they are not a small share of NHS spend. Out-patient use is a four-band count, so its intensive margin is crude. Elective and emergency admissions are indistinguishable, though they differ in cost by more than a factor of two. And every quantity is self-reported over a twelve-month recall window, against wave spacing that is not exactly twelve months.

The proxy is therefore a *secondary-care bed-day-weighted cost index*, and should be named that way in anything we write rather than called a cost.

7 The decision this sets up

Stages 3 and 4 have now run: concentration reproduces, the age profile does not, the diagnosis is flat unit costs rather than the banded instrument, and the convexity conclusion is stable across every costing assumption while remaining sensitive to the health scale. Given that the convexity survives the spell and outlier assumptions, then UKHLS alone supports the framework’s empirical claim and linkage is a refinement rather than a precondition.

If instead the shape moves with the spell assumption, or the proxy misses the published profiles, then the case for UK LLC linkage becomes concrete and specific — and we would know exactly which quantity we needed it for, which is a much stronger application than asking for HES in general. The binding risk there remains sample size: roughly 8,000 consented participants against the 84,679 in our current sample, with an extension to all participants stated but needing verification.

Effort

Three to five days end to end, entirely on data already on disk, with no new permissions. Stages 1 and 3 are the ones worth doing even if we stop there: a validated flat-rate index would already replace the physical-units analysis of §8 with something denominated in pounds.